

Assignment Title: (CO2-PO3)

Design and Performance Evaluation of Fundamental Data Structures

The objectives of this assignment are to:

1. **Design and implement a linked list data structure** to model patients residing on different floors of a hospital.
2. **Apply fundamental linked list operations** such as rearranging, sorting, searching, insertion, deletion, and traversal using both **manual (hand)** and **programmatic (code)** methods.
3. **Assess the performance and suitability** of linked lists for handling dynamic and priority-based data in a real-world application.
4. **Develop algorithmic and problem-solving skills** by mapping real-life constraints to appropriate data structure solutions.
5. **Achieve CO2** by designing and implementing an efficient data structure solution aligned with **PO3 (Design / Development of Solutions)** at the **C3 (Apply)** level.

Task 1: Application Scenario – Hospital Patient Management Using Linked Lists

Consider a hospital building with **20 floors**, where **one patient resides on each floor**. Each patient is identified by a unique number from **1 to 20**. As a doctor, you must visit every patient daily. Patients with the **smallest odd-numbered IDs are considered more critical** and should be visited earlier, while **even-numbered patients are considered normal**, though they must still be visited regularly.

Based on this scenario, perform the following tasks using a **Linked List data structure**:

Tasks:

1. **Design and create a linked list** representing the 20 patients according to the given scenario.
2. **Rearrange the linked list** so that all **odd-numbered (critical) patients appear before even-numbered (normal) patients**, using:
 - (i) Manual (hand) method
 - (ii) Programmatic (code) method
3. **Sort the linked list** in ascending order using both:
 - (i) Hand method
 - (ii) Code implementation
4. **Calculate the sum** of all patient node values using:
 - (i) Hand method
 - (ii) Code implementation
5. **Identify the middle patient** (or nearest middle patient if applicable) using:
 - (i) Hand method
 - (ii) Code implementation
6. **Remove all normal patients** from the linked list using:
 - (i) Hand method
 - (ii) Code implementation
7. **Insert a new highest-priority critical patient** into the linked list using:
 - (i) Hand method
 - (ii) Code implementation
8. **Search for a critical patient whose adjacent patient is normal**, using:
 - (i) Hand method
 - (ii) Code implementation
9. **Convert the singly linked list into a doubly linked list**, adding a **new patient as the head node**, using:
 - (i) Hand method
 - (ii) Code implementation

Analytical Questions:

1. Explain why the **linked list data structure** is particularly suitable for this hospital patient management system.
2. Discuss how selecting an **inappropriate data structure** could negatively impact system performance and resource utilization.

Guidelines:

Students must submit a **printed hard copy** of the assignment within the given deadline. The submission should be neatly organized, properly formatted, and complete. Late or soft-copy submissions will not be accepted.

Submission Deadline: 04.01.2026